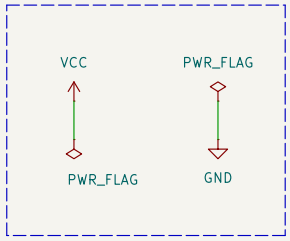
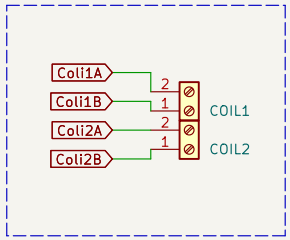
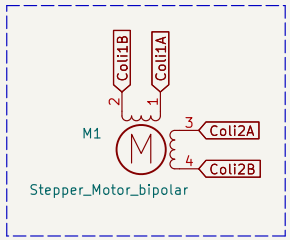
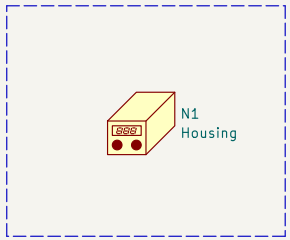
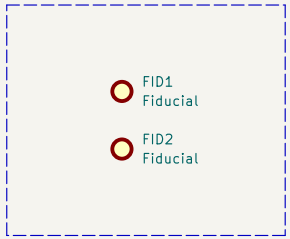
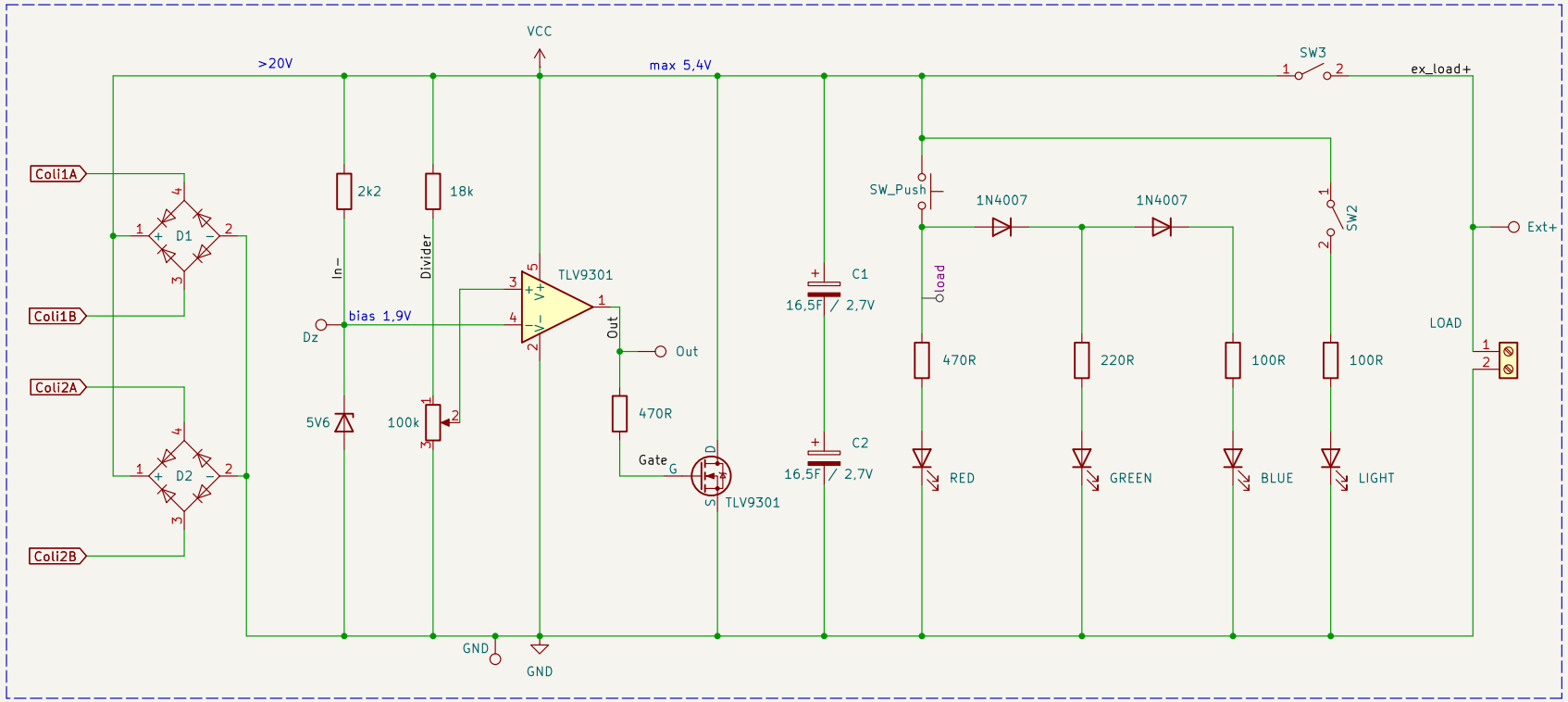










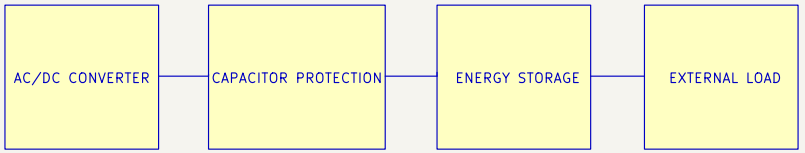


BILL OF MATERIALS:
"Reference,Value,Name"

SMD:
D1,MB10s_Bridge,Gretz_bridge,
D2,MB10s_Bridge,Gretz_bridge,
D3,5V6,Zener_Diode,
U1,TLV9301,Low-Dropout_Regulator,
R1,2k2,Resistor_1206,
R2,18k,Resistor_1206,
R3,470R,Resistor_1206,
R4,470R,Resistor_1206,
R5,220R,Resistor_1206,
R6,100R,Resistor_1206,
R7,100R,Resistor_1206,
SW1,SW_Push,Button_Switch

THT:
C1,16,5F/2,7V,Capacitor,
C2,16,5F/2,7V,Capacitor,
COIL1,Screw_Terminal_01x02,
COIL2,Screw_Terminal_01x02,
ext1,LOAD,Screw_Terminal_01x02,
D4,RED,LED_THT:LED_D3.0mm,
RV1,100k,Potentiometer,
Q1,IRLZ44N,NMOS,
SW2,Button_Switch,
SW3,Button_Switch,
D6,GREEN,LED_THT:LED_D3.0mm,
D8,BLUE,LED_THT:LED_D3.0mm,
D9,LIGHT,LED_THT:LED_D3.0mm,
D5,1N4007,Diode_SMD,
D7,1N4007,Diode_SMD,

Exclude from
BOM:
M1,Stepper_Motor_bipolar,
N1,Housing



Included:
Main schematic,Block diagram, Bill of materials, Excluded components from BOM
Owned by: G.F.W

Sheet: /
File: crank_gen_v.0.4_pcbway-template.kicad_sch

Title: Off-grid Hand crank generator

Size: A4	Date: 12/07/2026	**Rev: v.0.4**
KiCad E.D.A. 10.0.4		Id: 1/1